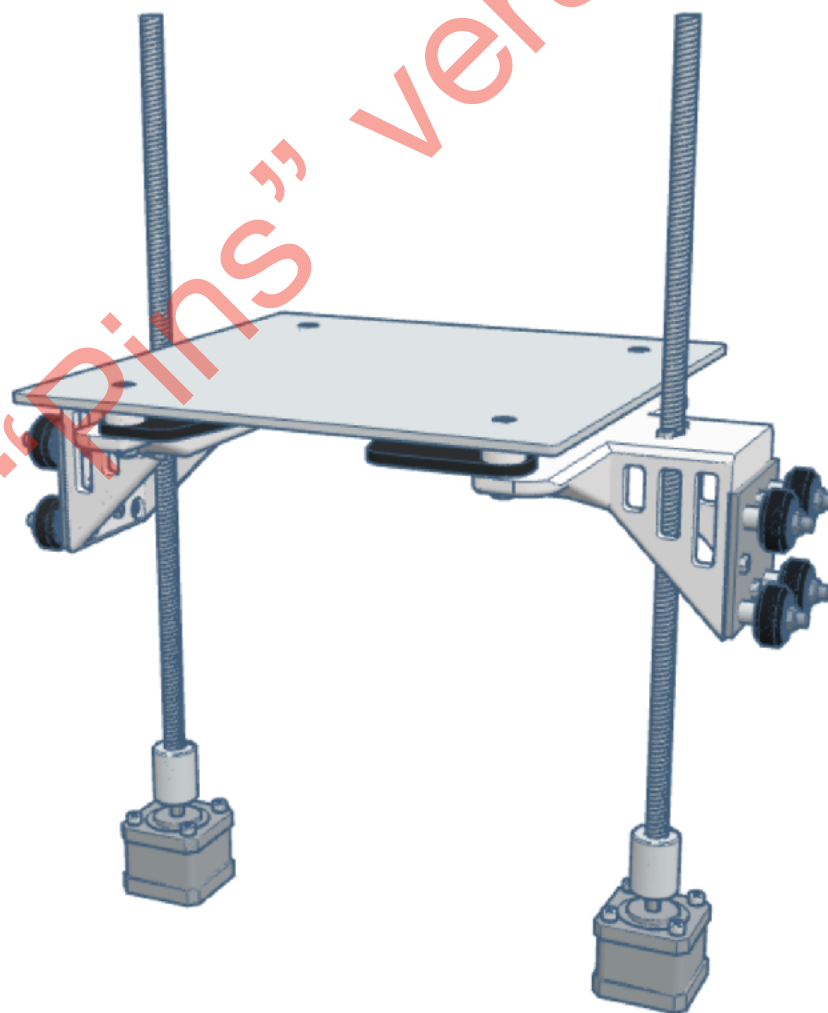




Duender
V-wheels Z axis

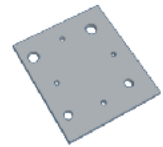


Your Open-Source 2 x Creality Ender-3 core XY conversion

by Sergei Irbis

Assembling the Z axis Z gantry

16



Open Build plates

2



M3x16 BHCS

4



M3x8 BHCS

16



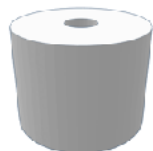
M4 Nylstop nut

4



M3 Nylstop nut

12



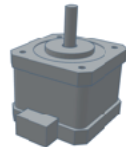
12mm silicone spacers

4



M4x35 FHCS
(from Ender 3)

2



Nema 17 42-34
(from Ender 3)

4



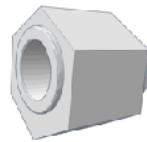
M5x30 BHCS
(from Ender 3)

16



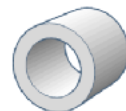
M5 Nylstop nut
(from Ender 3)

4



Eccentric M5 nut
(from Ender 3)

12



Wheel M5 spacer
(from Ender 3)

4

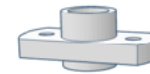
4



POM V-Wheel
(from Ender 3)

8

2



Lead screw nut
(from Ender 3)

2

8



8mm lead screw
(from Ender 3)

2

8



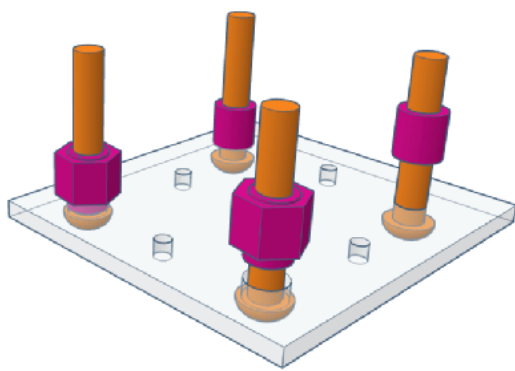
Motor shaft coupler
(from Ender 3)

2

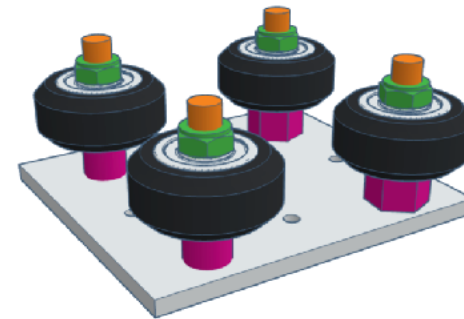
4

Assembling the Z axis

17

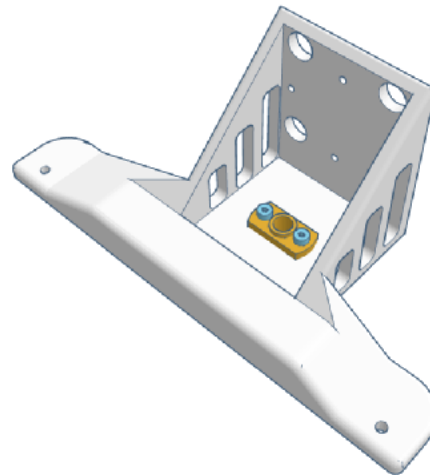


Insert 4 M5x30 BHCS from the underside
Spacers go over the smaller holes and
eccentric nuts over the larger ones
Those are all from the original Ender 3 roller
mounts



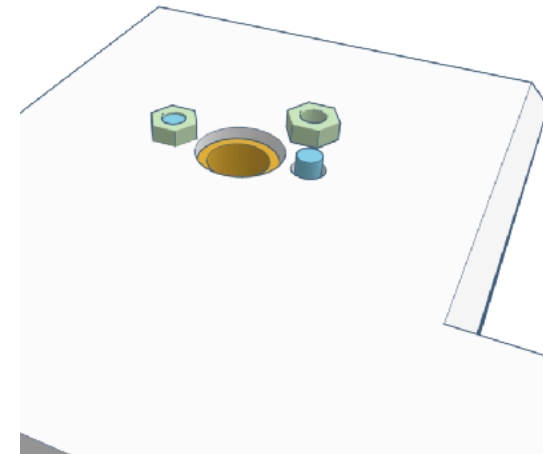
Finally, add 4 V-rollers from your Ender 3,
and an M5 Nylstop (self-locking) nut

**DO NOT TIGHTEN THE NUTS YET, you
need the whole screws/wheels to be loose**



Install the lead screw nuts under the plastic bed holders,
and insert 2 M3x16 SHCS or BHCS (Or you can use the
ones originally coming from your Ender 3)

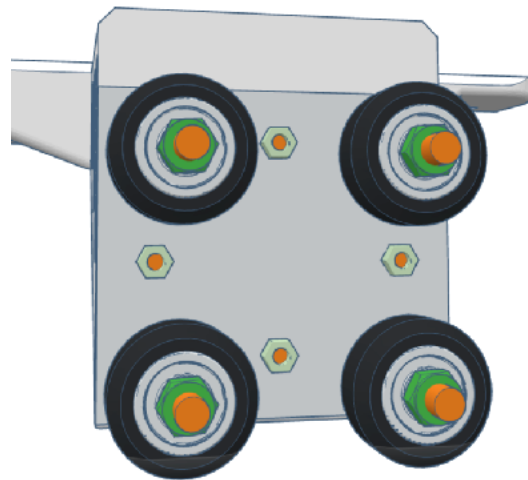
**If your lead screws aren't perfectly straight, it is
advised that you swap the original lead screw nuts
from the Ender 3 with some anti-backlash ones, to
avoid Z wobble**



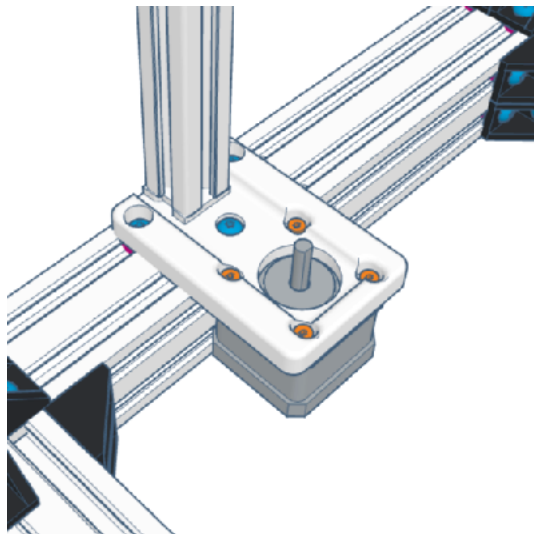
Turn the bed holders around being careful not to
lose the screws and secure them with M3 nuts

Assembling the Z axis

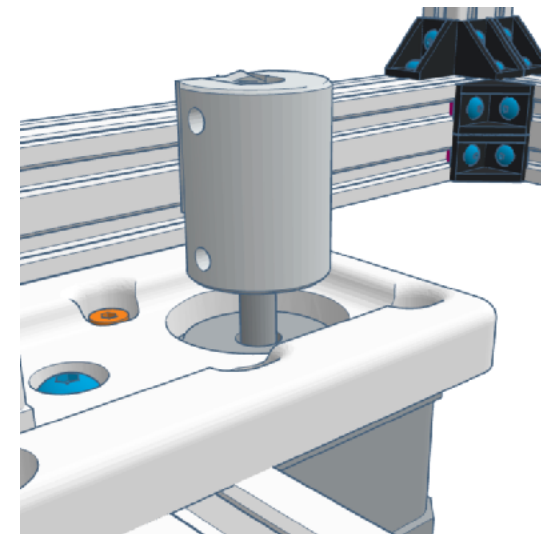
18



Place the roller plate behind the bed holder and use 4 M3x8 BHCS to hold it
Secure the screws with 4 M3 nuts



Place the Z motors under their plates and use 4 M3x8 BHCS to secure them. Orient the motor plug to the back of the printer

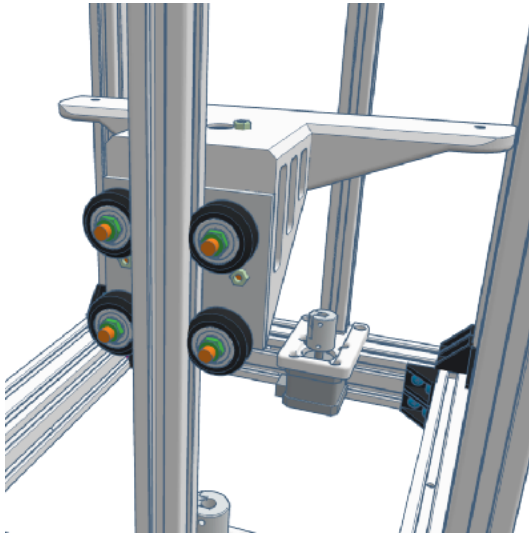


Install the lead screw shaft couplers onto the Z motor shafts and tighten only the lower screw

If your lead screws aren't perfectly straight, you may want to replace those with "spring" or "elastic" ones to avoid Z-wobble

Assembling the Z axis

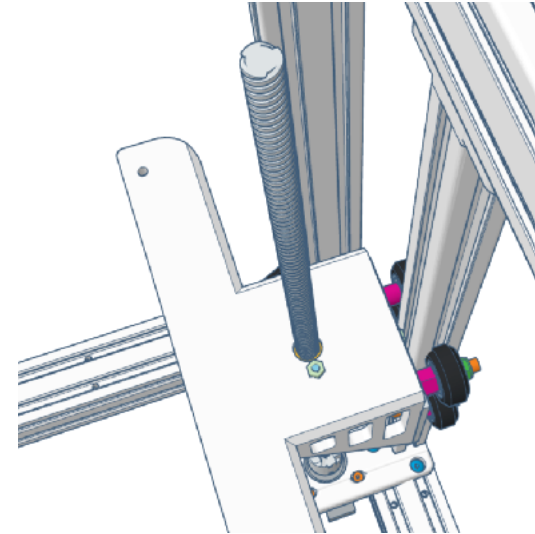
19



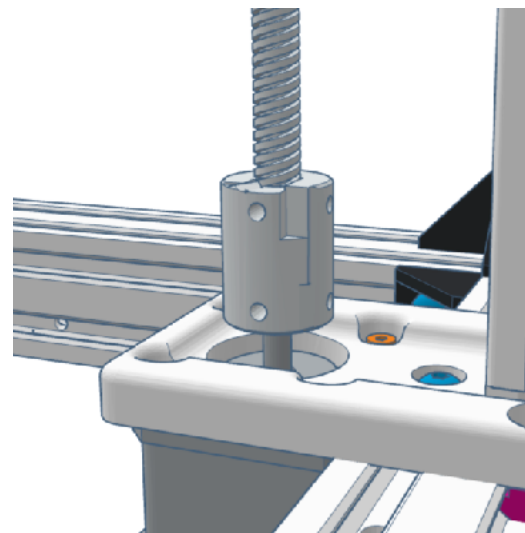
Insert the bed holder assembly onto the Z axis (note that the longer side goes towards the front) and tighten the V-wheels
Once secured, adjust the eccentric nuts to ensure all 4 wheels apply equal pressure

The wheels should rotate smoothly with a slight resistance when turned by hand

Double-check that the screw heads don't touch the extrusion surface.



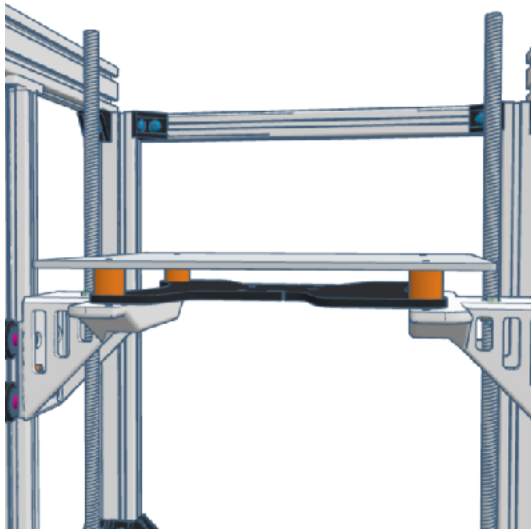
Insert each lead screw into its lead screw nut from the top then insert them into the Z motor coupler by rotating them until fully seated



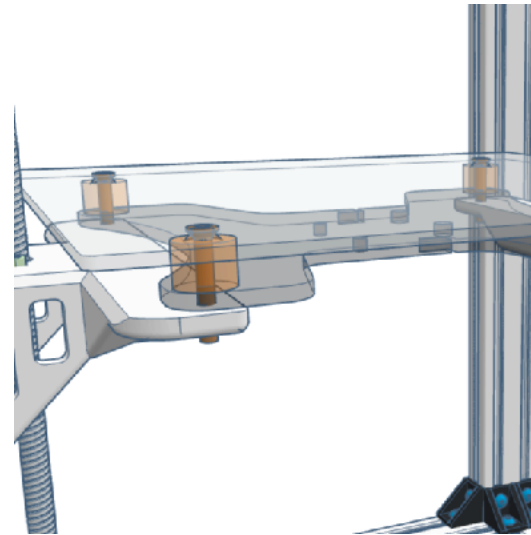
Tighten the upper screw of the coupler to hold the lead screw

Assembling the Z axis

20



Place the Ender 3 Y plate onto the bed holders, followed by 4 silicone spacers (12mm version shown here in orange) or you may use the springs from the Ender 3, then the heatbed. Of course, the wiring of the heatbed should end towards the back of your printer.



Use the original 4 M4x35 FHCS and secure them under the bed holders with M4 nuts.

